

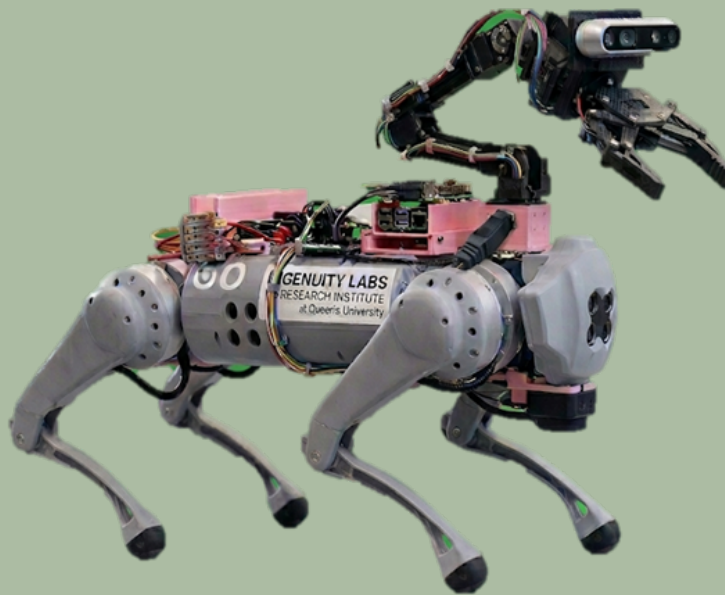


INGENUITY LABS
RESEARCH INSTITUTE
at Queen's University



SMITH
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MREN403 - MECHATRONICS AND ROBOTICS DESIGN IV



Final Design Report

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April 15, 2026



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AUTONOMOUS SMALL WASTE COLLECTION ROBOT

1 Introduction and Motivation

Improper disposal of litter is a persistent problem in urban areas. Beyond diminishing the visual appeal of shared spaces, waste can leach toxins into soil and water, harm wildlife, and pose health risks to humans. In nearly all modern cities, waste and other pollutants can be found in public areas and other spaces (see Figure 1). In fact, correspondence with the City of Kingston revealed a recent spike in illegal disposal of household waste by citizens to avoid the rising costs of municipal garbage bag tags [1].



Figure 1: The *Autonomous Small Waste Collection Robot* during a field trial at Summerhill, April 2026. Many cans, bottles and other waste items can be seen scattered across the ground, polluting the environment.

Litter comes from various sources, including pedestrians, motorists, nearby households, and waste bins overturned by weather or animals. Traditional collection relies on municipal workers, organizations, and volunteers, yet manual cleanup is tedious, repetitive, and sometimes hazardous [2].

Autonomous litter collection offers a valuable complement to existing initiatives. Robots can perform scheduled cleanups, reduce human exposure to unsafe conditions (e.g., roadside or unsanitary waste), and operate where litter accumulates rapidly.

Recent research has explored robotic litter collection with varying degrees of success. For example, [3] demonstrated a deep learning-based detection system coupled with a manipulator for pick-and-place collection. However, this system was mounted on a stationary platform and struggled in outdoor environments where lighting, weather, and background complexity degrade

performance. AutOTranS [4] extended litter detection to outdoor settings, but under limited scenario variability, while [5] used a similar robotic platform but enhanced detection robustness and integrated tactile-feedback for improved grasping.

Building on these foundations, this project aims to advance autonomous litter collection by addressing both perception and collection in unstructured environments.

2 Requirements and Specifications

The *Autonomous Small Waste Collection Robot* project aims to design hardware and software architecture for a robot to autonomously detect and collect litter (i.e., garbage, waste, foreign objects) in urban settings, specifically around the Queen’s University campus and the City of Kingston’s downtown core.

2.1 Stakeholders

Six key stakeholders influenced the project: Dr. Jonathan Gammell, the City of Kingston, Queen’s University, Ingenuity Labs, the general public, and the environment. Their interests were classified into two groups: those affected by the robot’s deployment (environment, City of Kingston, Queen’s, and public) and those invested in its development (Dr. Gammell, Queen’s, Ingenuity Labs). The former are concerned with compliance, safety, and property protection, while the latter provide supervision, resources, and platforms in support of research and development. Notably, the collaboration established between the developing team and Ingenuity Labs equipped the team with access to additional resources in exchange for research demonstrations and outreach content.

2.2 Context and Scope

The scope of this project encompassed the design and implementation of an autonomous robotic system capable of localizing and collecting litter in outdoor environments, with a primary focus on the physical collection process.

The project architecture specifically focused on integrating three core subsystems:

- A detection model designed to classify waste types commonly found in urban spaces, prioritizing broad class coverage and robust model confidence.
- A manipulation system with a trajectory-planning strategy capable of handling waste items of varying geometries and orientations.
- Integration of the manipulation system onto a mobile quadruped chassis with a robust navigation stack to enable operation through diverse environments.

Given the accelerated seven-month development timeline, the system was developed as a proof-of-concept to demonstrate core functional capabilities and improvements over prior solutions, rather than as a fully mature commercial product.

To ensure the successful integration and demonstration of these subsystems, the operational environment was constrained to controlled, semi-structured *indoor* pathways (e.g., hard flat surfaces without dynamic obstacles or poor weather) (**OP-01**, **OP-02**, **OP-04**). Furthermore, the target waste was limited to stationary cans and bottles (**CV-01**), a prominent subset of litter observed in the Queen’s University community and capable of being grasped and lifted by the selected manipulator (**ME-01**).

A comprehensive breakdown of all functional requirements, safety protocols, physical constraints, and limitations, is documented in Appendix A.

3 System Architecture & Design

The autonomous litter-collecting robot integrates several subsystems, each supporting a stage of the collection process. The project strategy outlines how these subsystems interact to form a cohesive system. Central to the design is the Collection Strategy (3.2), a five-step procedure guiding how the robot identifies and collects litter. Each subsystem plays a specialized role: the Computer Vision module (4.1) detects and classifies litter, providing target positions to the control software; the Manipulation system (4.3) enables precise grasping; and the Mobile Platform (4.4) supplies locomotion, onboard compute, and sensor fusion for autonomous navigation.

3.1 Complete Hardware System

The complete hardware system is comprised of multiple components. The OpenMANIPULATOR-X arm by ROBOTIS is fixed with an Intel RealSense D435 camera. This setup is mounted at the front of the Unitree Go1 robot as a mobile base. An RPLidar C1 and MPU-6050 Inertial Measuring Unit (IMU) are fitted to the bottom of the robot as well.

The system is driven by a 3S LiPo battery providing power to a Raspberry Pi4, Raspberry Pi5, and subsequent components. A complete bill of materials is provided in Table 3 of Appendix B. Further discussion of the electrical setup is provided in Section 3.1.3.

3.1.1 Go1 Quadruped Robot

Through the collaboration pact with Ingenuity Labs, the team has access to a Unitree Go1 Edu[6], a lightweight quadruped robot platform. See Figure 2

Legged robots have high agility over uneven terrain (e.g., rocks, hills, and grass) and their discrete padded footsteps minimize soil impact compared to wheeled differential-drive vehicles. In addition, legged platforms present a more natural, zoomorphic appearance which may be perceived as less intimidating and less intrusive in public settings.

To mitigate risks of damaging or corrupting this loaned equipment, we opted not to develop on internal components, including the onboard Nvidia Jetson and Raspberry Pi. As a result, there was no feedback from the joint encoders or internal IMU for pose estimation. This leads into developing an external component package.

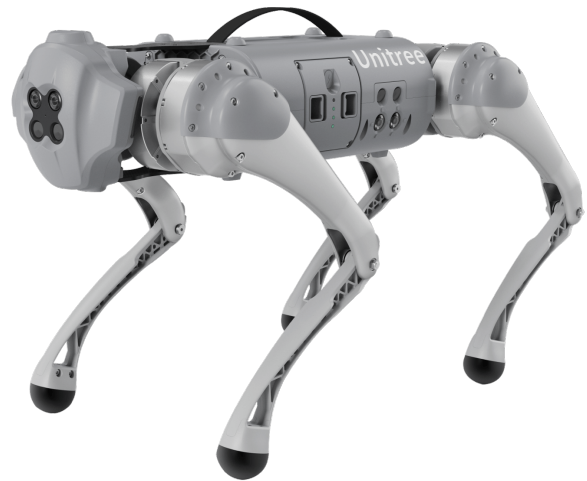


Figure 2: Unitree Go1 Edu, an intelligent legged robot (Source: Unitree Robotics [6]).

3.1.2 Sensors and Components

C1 RPLidar

The RPLidar C1 is a compact, high-performance 2D laser scanner developed by SLAMTEC. It operates by emitting an infrared laser beam that reflects off surrounding objects; by measuring the time of flight or phase shift of the reflected light, the sensor calculates precise distances. It provides a 360-degree omnidirectional scan of the environment and a 16 metre detection radius [7].

MPU6050 IMU

The MPU6050 combines a 3-axis gyroscope and a 3-axis accelerometer into one component. The accelerometer measures linear acceleration, while the gyroscope measures angular velocity. These two sensors create the external pose estimation feedback for the Go1 platform. More information is presented in Navigation: 4.4.

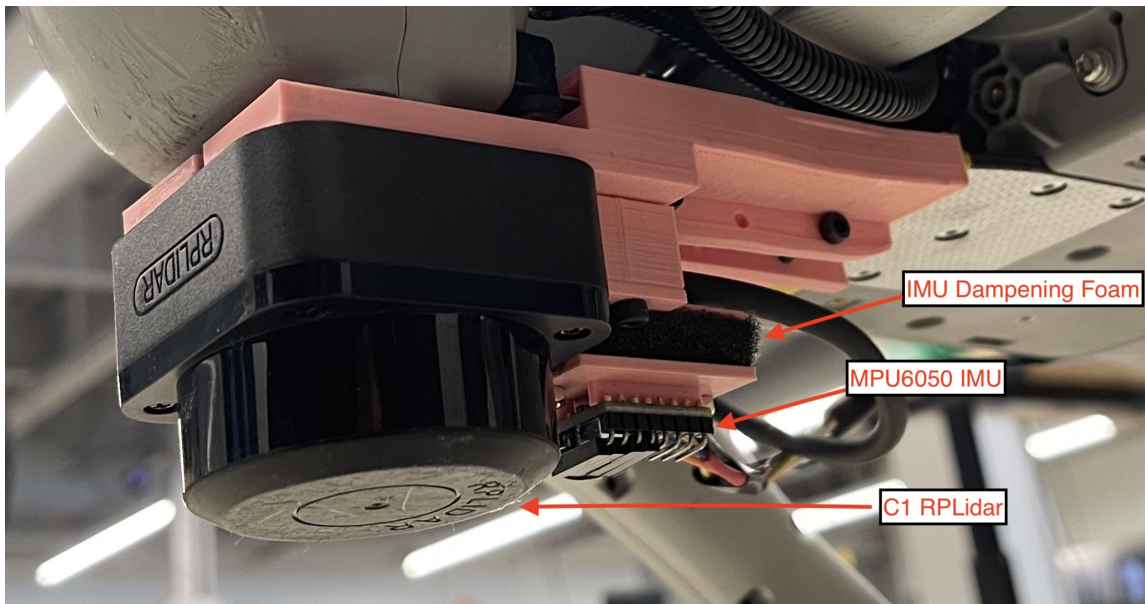


Figure 3: C1 RPLidar and IMU Combination.

OpenMANIPULATOR-X

The OpenMANIPULATOR-X robot arm is an open-source platform by ROBOTIS. It is a four degree of freedom serial manipulator, with an additional motor to open and close the gripper. The joints are made of Dynamixel XM430-W350 T motors and can carry a payload up to 500 g and the manipulator has a maximum reach of 380 mm[8]. Dimensions of the manipulator can be seen in Figure 4.

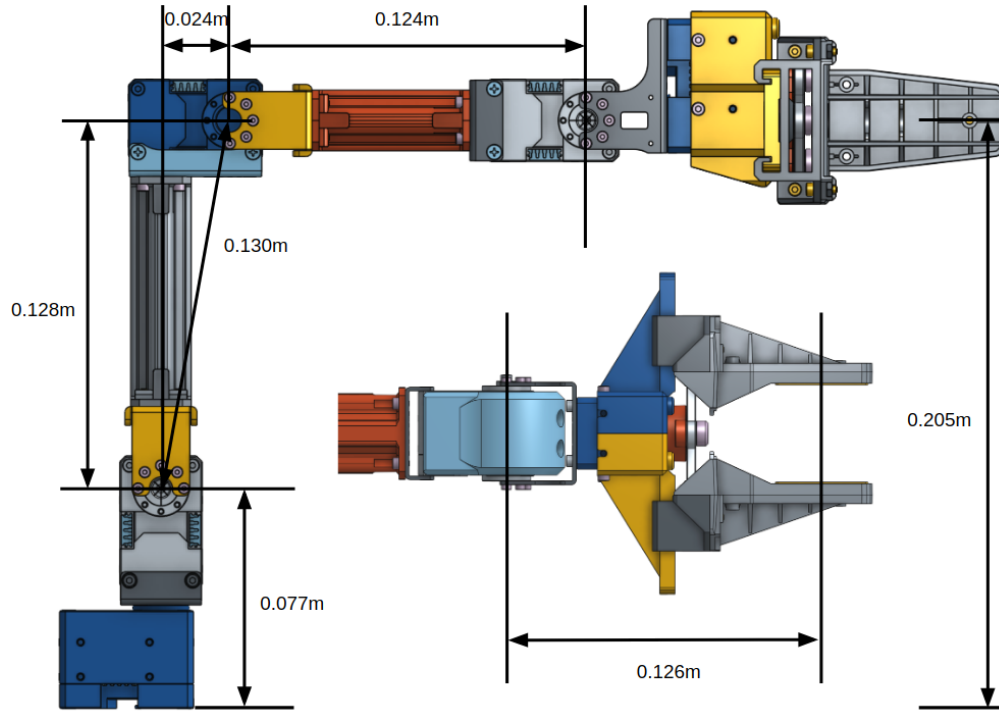


Figure 4: Geometric specifications of the OpenMANIPULATOR-X arm (Source: ROBOTIS [8]).

Intel RealSense D435

The Intel RealSense D435 is a stereo depth camera. It features a pair of infrared sensors, an RGB sensor, and an infrared projector, allowing it to calculate depth by measuring the disparity between images. In this implementation, the D435 camera is utilized for computer vision, specifically for can¹ detection [9].



Figure 5: Intel RealSense D435 Camera [9].

3.1.3 Electrical Setup

An electrical diagram of the main components and their connections is seen in Figure 6. A more detailed diagram explaining the connections between each component is presented in Figure 7 in Section 3.2.

¹Without loss of generality, this report often refers to a detection or target object as a “can” for simplicity, though the detected object could very well be of another class within our target set described in Section 2.2.

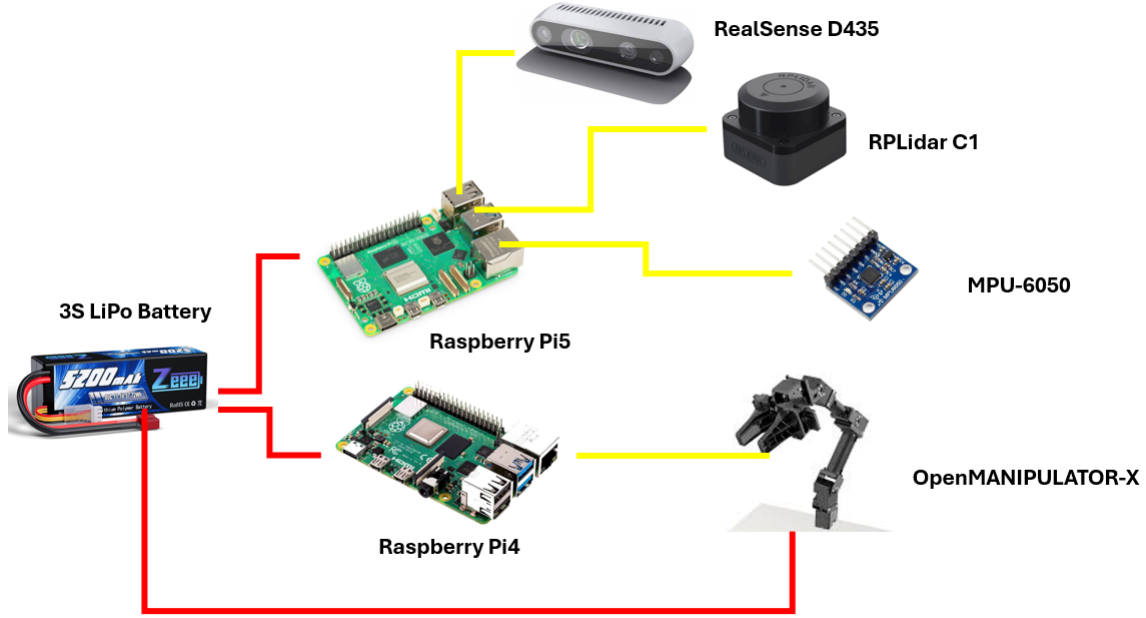


Figure 6: Basic electrical diagram showing connections between main components of the hardware system, save power converters and power distribution/control boards. *Red lines signify power, yellow lines signify control signals.*

3.2 Collection Strategy

Figure 7 details the hardware and software architecture and information flow. Utilizing the computational nodes described in Figure 7, the autonomous collection system executes a five-state procedure to efficiently navigate and retrieve litter, as illustrated in Figure 8.

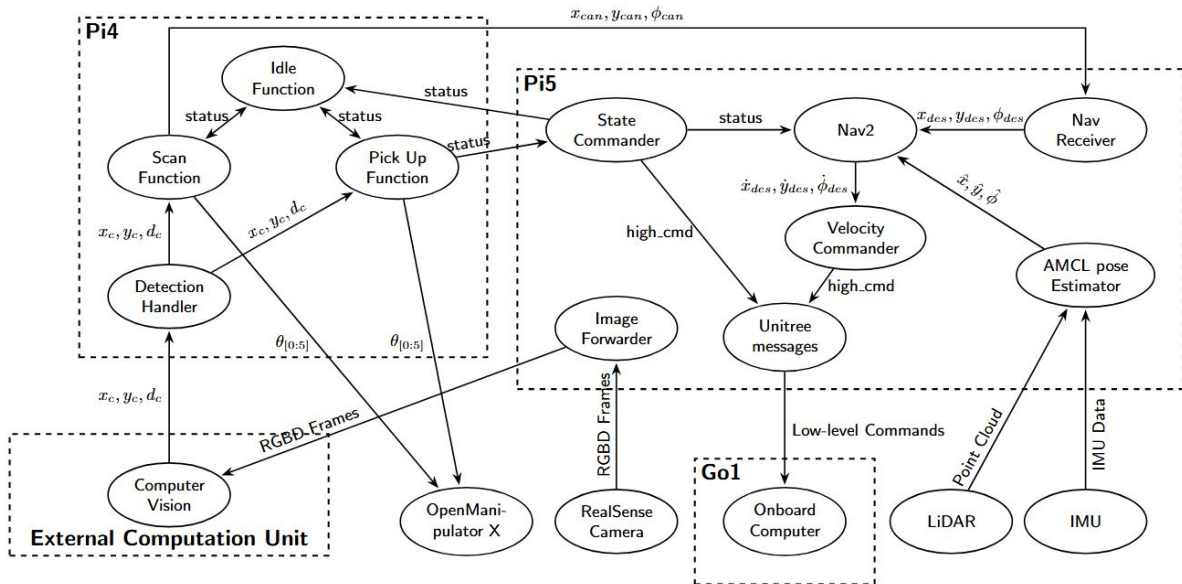


Figure 7: The robotic system architecture is distributed across four computation units (dashed boxes) and four sensor and actuator units. Computation units communicate over a shared WiFi network using UDP and TCP messages, while sensor and actuators are accessed via USB connections.

1. **Mapping & Localization** Upon initialization, the robot must localize itself within the global map based on a prior belief, LiDAR observations, and IMU updates.
2. **Search:** The robot navigates the environment while simultaneously using its onboard cameras to detect potential litter. The *scan function* pans the manipulator searching for trash.
3. **Goal Transform:** The camera feeds are processed by a *computer vision* model to locate and classify litter, then converts the detected object's camera coordinates to the manipulator's base frame defining a target location for the robot to approach. .
4. **Approach:** *Nav2* coordinates the robot's transition, generating velocity commands and forwarding them to *Unitree Messages*. *State Commander* sets the robot's state crouch and dampens motors.
5. **Pick and Place:** *Pick Up Function* instructs the manipulator to pick up the target using camera feedback. The end-effector is oriented optimally to securely grasp the litter item. The robot deposits the collected litter in a bin mounted on top of its chassis.

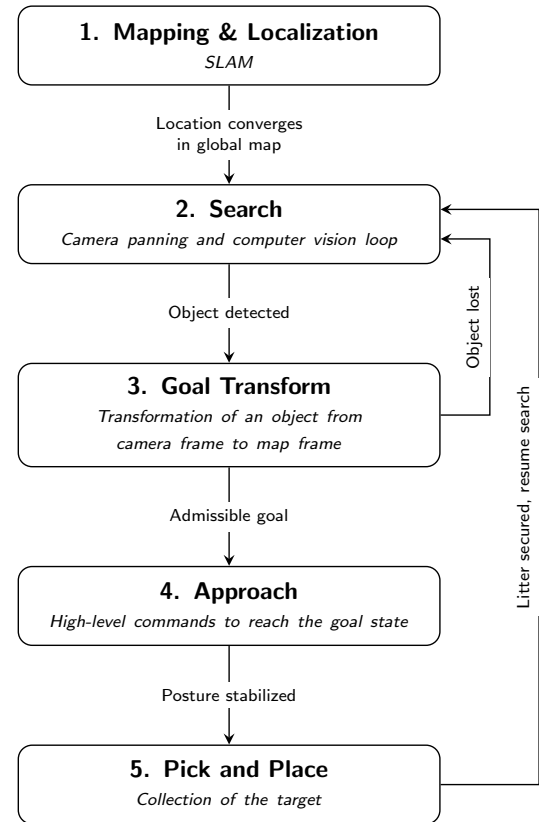


Figure 8: Simple flow diagram of the waste collection strategy.

4 Detailed Implementation

4.1 Computer Vision

Model

The model used in the project is a retrained YOLOv8 instance segmentation model. YOLO (You Only Look Once), developed and maintained by Ultralytics, was selected for several reasons. The architecture has an extensive development history and a large active community, signifying documentation, pretrained weights, and community support are readily available. The Ultralytics training framework abstracts much of the complexity of the training pipeline, allowing rapid iteration of hyperparameters and augmentation strategies without requiring low-level framework modifications. Additionally, YOLOv8 supports instance segmentation natively, which is necessary for extracting per-object mask geometry used downstream in the pipeline. The specific variant selected is YOLOv8s-seg (small), which is one step up from the smallest available variant, YOLOv8n-seg (nano). This choice reflects the primary deployment constraint: inference must run at a practical frame rate on hardware with limited compute. The small variant provides a meaningful improvement in model capacity over the nano variant while remaining practical on the hardware. The model was trained separately on a personal computer with a discrete GPU and then deployed onto the mobile platform.

Data

Training data was sourced from Roboflow, where four publicly available labeled datasets covering metal can and plastic bottle instance segmentation were obtained: one can segmentation dataset and three bottle segmentation datasets. The datasets contain polygon-annotated images with varied backgrounds across both indoor and outdoor environments, and include a range of object types beyond standard beverage containers, such as soup and tuna cans. In total, 4283 images were used across training, validation and test sets.

Once downloaded, pre-processing was required before training. Each dataset used different class naming conventions, so class names were normalized to a unified two-class schema: can and bottle. Aliases such as c, b, bottles, bottle_pleine, and bottle_vide were remapped accordingly. Annotations without valid segmentation polygons were discarded, retaining only entries with full polygon masks. The four datasets were then merged and reshuffled, and a unified 80/10/10 train/validation/test split was applied, replacing the original per-dataset splits to ensure a consistent and unbiased distribution.

Training

After data pre-processing, the model was trained on a CUDA-enabled NVIDIA RTX 3080 Ti GPU for up to 300 epochs with an early stopping patience of 20 epochs, halting if validation performance fails to improve. A batch size of 16 and an input resolution of 640×640 pixels were used. A fairly aggressive augmentation pipeline was applied to improve generalization, given the relatively limited training data available. Geometric augmentations included random rotation up to ± 90 , scaling up to 80% of the original image size, shearing at ± 10 , translation by up to 10% of image dimensions, vertical and horizontal flipping at probabilities of 0.1 and 0.5 respectively, and a perspective distortion factor of 0.0005. Photometric enhancements targeted hue ($\pm 1.5\%$), saturation ($\pm 70\%$) and value ($\pm 40\%$) in HSV color space. Mosaic augmentation (probability 1.0) and Mix-up blending (probability 0.1) were also enabled to expose the model to composite training scenes. Quantitative training results are discussed in Section 5.3.

Inference

After training, the model processes the video feed from cameras onboard the mobile platform to produce segmentation masks around detected pieces of litter. In conjunction with OpenCV, properties of each detected object are extracted by fitting an ellipse to the segmentation mask, yielding the major and minor axis lengths and the orientation angle relative to vertical. Using the onboard RealSense RGB-D camera, the depth at the centroid of each mask is sampled to convert the axis dimensions from pixels into metric units. This information is passed to the guidance and manipulator subsystems, which plan a path to each detected piece and direct the manipulator accordingly. Due to the computational limitations of the Raspberry Pi, the final version of the vision pipeline incorporates an external computer, via a TCP protocol from the Pi 5, and a UDP protocol to the Pi 4 with the detected object information using dedicated sender and receiver nodes.

4.2 Goal Transformations

To convert a CV detection into an admissible goal for the navigation system we perform a series of transformations with several simplifying assumptions. First, we center the object within the camera frame using a visual-servoing PID control of the manipulator. The computer vision

model provides pixel coordinates, x_{pixel} , y_{pixel} and depth d_C . This centering reduces error in the depth measurement and if x_{pixel} , y_{pixel} are centered in the camera frame, we can assume the physical camera's x -axis points directly to the object. Under this assumption the detected can can now be described in a cartesian vector-space and we can apply common vector operations. Figure 9 shows simplified 2D diagrams of some of these transformations. Subscript B denotes the base-link frame which we wish to express the can location in, subscripts EE and C denote the manipulator's end-effector and camera frames respectively. Let the position and rotation of the end-effector relative to it's base-link be denoted by $\mathbf{p}_{EE} \in \mathbb{R}^3$ and $\mathbf{R}_{EE} \in SO(3)$, accessible via the OpenMANIPULATOR-X API. The camera is offset from the EE by a translation $\mathbf{t}_C^{EE}$ and a pitch rotation $\mathbf{R}_C^{EE}$ defined by a tilt angle $\beta = 30^\circ$:

$$\mathbf{t}_C^{EE} = \begin{bmatrix} -0.025 \\ 0.0 \\ 0.03 \end{bmatrix}, \quad \mathbf{R}_C^{EE} = \begin{bmatrix} \cos \beta & 0 & \sin \beta \\ 0 & 1 & 0 \\ -\sin \beta & 0 & \cos \beta \end{bmatrix}$$

The global position of the camera center $\mathbf{p}_C$ and its optical forward-pointing unit vector $\mathbf{x}_C$ are calculated by transforming the local offsets into the world frame:

$$\mathbf{p}_C = \mathbf{p}_{EE} + \mathbf{R}_{EE} \mathbf{t}_C^{EE}$$

$$\mathbf{x}_C = \mathbf{R}_{EE} \mathbf{R}_C^{EE} \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^T$$

Given a depth measurement d_C representing the distance to the waste object, the initial global coordinates of the object $\mathbf{p}_{\text{can}}$ are found via linear projection:

$$\mathbf{p}_{\text{can}} = \mathbf{p}_C + d_C \mathbf{x}_C$$

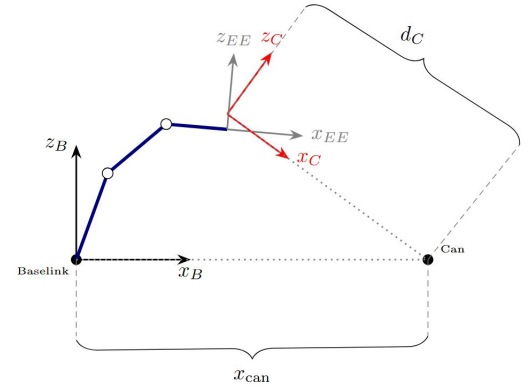
Figure 9 (a) shows a case of this projection in the baselink's x - z plane. As the robot's map is a 2-D workspace, we apply the assumption that the z component of the can location may be discarded without loss of information and we only consider $(x_{\text{can}}, y_{\text{can}})$. To facilitate the pickup we assume it is optimal to align the robot's heading with the can's major axis. The major axis calculated by the CV algorithm θ_C is assumed to be equal to the separation angle of the can's true major-axis from the direction vector $(x_{\text{can}}, y_{\text{can}}, 0)$, θ as shown in Figure 9 (b). A heading angle α is computed using the object's Cartesian coordinates and the measured major axis θ_C :

$$\alpha = \phi + \theta_C$$

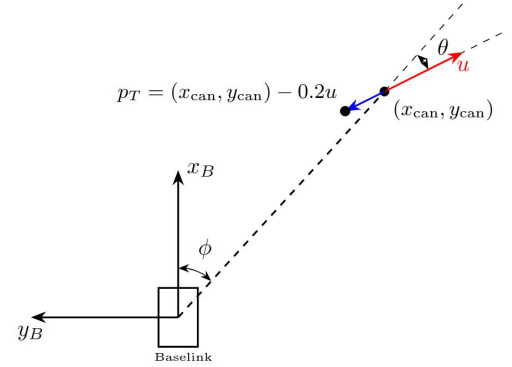
where $\phi = \text{atan2}(y_{\text{can}}, x_{\text{can}})$.

The target position $\mathbf{p}_T$ includes a 0.2 m standoff along the heading vector $\mathbf{u} = [\cos \alpha, \sin \alpha]^T$ to center the object in the manipulator's workspace:

$$\mathbf{p}_T = (x_{\text{can}}, y_{\text{can}}) - 0.2\mathbf{u} \quad (1)$$



(a) Side-view projection of the goal transform.



(b) Top view showing calculation of goal offset and heading.

Figure 9: Idealized transformations convert raw CV detections in camera frame C into a 2D navigation goal with a defined standoff distance and orientation in the manipulator's baselink frame, B . The subfigures show these transformations in simplified planar cases.

The tuple $\mathbf{p}_{target} = (x_T, y_T, \alpha)$ relative to the manipulator's baselink must next be transformed to the map global map space. For Nav2's *Velocity Commander*, an admissible goal consists of a position, p_{goal} and orientation expressed as a quaternion, $\mathbf{q}_{goal}$. In calculation of $\mathbf{q}_{goal}$, we can ignore the roll and pitch, simplifying the conversion to rotate strictly around the z-axis. These goals in the map frame are calculated as:

$$\mathbf{p}_{goal} = \begin{bmatrix} x_{goal} \\ y_{goal} \end{bmatrix} = \begin{bmatrix} x_R \\ y_R \end{bmatrix} + \begin{bmatrix} \cos \theta_R & -\sin \theta_R \\ \sin \theta_R & \cos \theta_R \end{bmatrix} \begin{bmatrix} x_T \\ y_T \end{bmatrix}, \quad \mathbf{q}_{goal} = \begin{bmatrix} 0 & 0 & \sin \frac{\theta_R + \alpha}{2} & \cos \frac{\theta_R + \alpha}{2} \end{bmatrix}^T \quad (2)$$

Where (x_R, y_R) is the robot's 2D pose estimate and θ_R is the robot's heading in the map space. Finally, the goal pair $(\mathbf{p}_{goal}, \mathbf{q}_{goal})$ is ready to be sent to the navigation controls described in Section 4.4.3.

4.3 Manipulation

4.3.1 Development Toolkit

The manipulator is controlled using ROS Noetic packages developed by the ROBOTIS team. These packages provide launch files that allow the programmer to manually move the manipulator via GUI or keyboard teleoperation. Custom services are also available, which allow the programmer to move the manipulator in joint space and task space using forward and inverse kinematics calculations respectively.

Using the given framework, a collection algorithm was developed to center and approach an object autonomously, using feedback from the camera, before using open-loop control to grab the object. A flowchart of the collection algorithm is detailed in Figure 10.

4.3.2 Algorithm Implementation

The scan and pickup algorithms were written in Python, leveraging services provided by ROBOTIS using the rospy library. Though simple, the control algorithms required handling of specific edge cases, as the camera would often lose the object of interest when the manipulator got too close or when environmental conditions did not match the ones on which the model was trained.

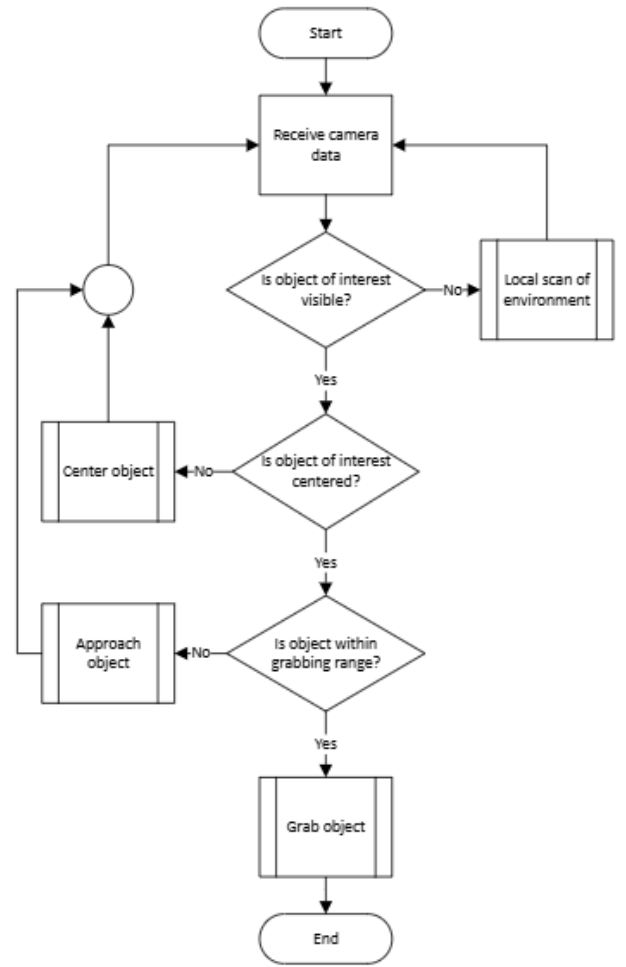


Figure 10: Flowchart detailing the collection algorithm of the manipulator subsystem.

This resulted in the development of a local search algorithm where the arm searches its local area to find the lost object and resume to closed-loop position control. Another feature was developed from the constraint of the arm's small reachable workspace. In the case that the centroid of the object would be too far to reach and center, the arm checks if any point on the can is reachable and proceeds accordingly. When the arm is picking up an object that results in full extension, it tilts the wrist joint to a 45 degree angle to further extend its reach to successfully grab the object.

4.3.3 Custom Camera Mount

As the camera was mounted on top of the gripper, the camera frame was centered on the area above the gripper instead of ideally, in the center of it. An example can be seen in Figure 11. To resolve this, a camera mount angled 30 degrees downwards from the horizontal axis of the gripper was designed and adopted by the robot. The angled camera resulted in a frame where an object centered below the gripper would be visible, allowing the manipulator to center in on the object accordingly.

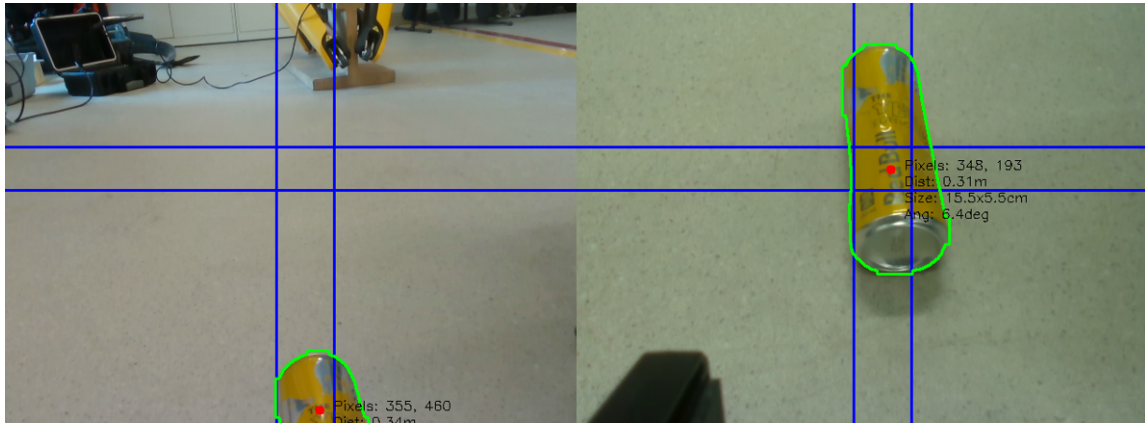


Figure 11: *Left:* A frame given by the RealSense camera mounted on the arm facing horizontal. *Right:* A frame given by the RealSense camera mounted on the arm at a 30 degree angle downwards from the horizontal.

4.4 Navigation

The navigation implementation for Go1 integrates synchronized hardware inputs with the ROS2 layer to achieve autonomous locomotion. This process relies on a multi-stage pipeline: robust state estimation through sensor fusion, the generation of a spatial occupancy grid via SLAM, and the execution of path-planning algorithms through Nav2. By establishing a reliable transform tree and precise communication protocols, the system is capable of complex tasks such as the autonomous approach of target objects.

4.4.1 EKF

The EKF predicts the robot's next state based on previous movement and then corrects that prediction using new sensor measurements, effectively weighting each sensor based on its perceived reliability to provide a smooth, accurate pose.

In order to localize in a map, a `map` → `odom` → `base_link` transform tree needs to exist. The `base_link` is the center frame of the robot. By using the pose estimation created from the EKF, an odometry topic in ROS 2 can be created, represented as `odom`.

4.4.2 SLAM

The SLAM library—Simultaneous Localization and Mapping—requires an odometry transform to transform the LiDAR points and create the map. By correlating LiDAR scans over time, SLAM identifies structural features to stitch together a coherent spatial representation. In other words, a map.

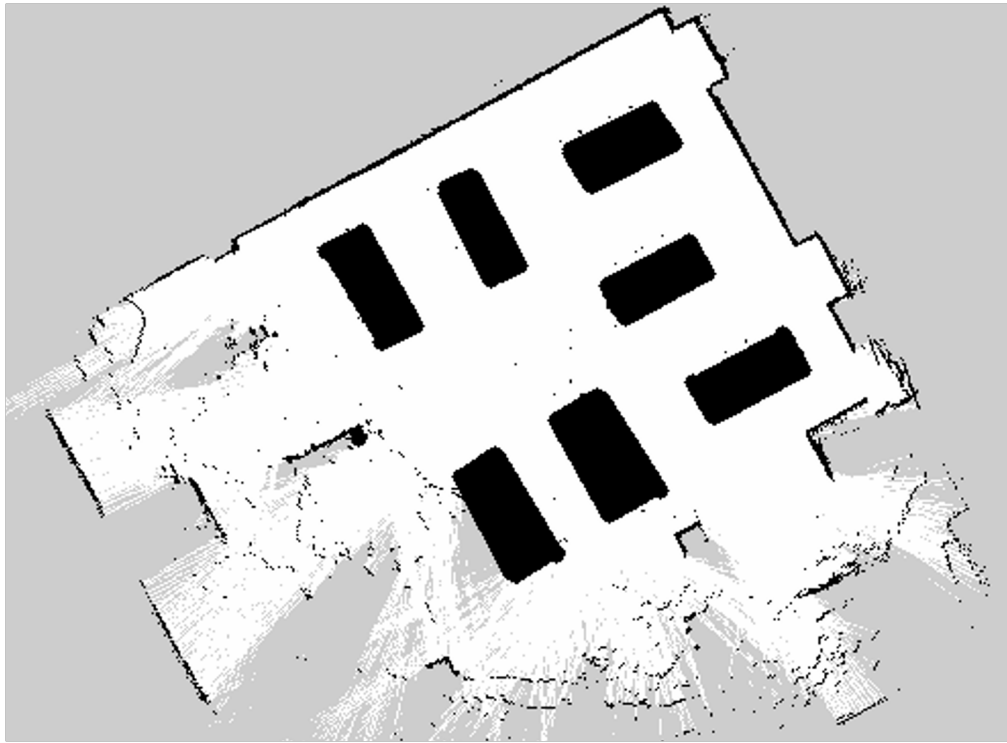


Figure 12: SLAM Map of Mitchell Hall Room 235–The Final Showcase Location

4.4.3 Nav2

Once a map is created, Nav2 (a navigation library provided by ROS2) is brought up with AMCL and the previously created SLAM map. AMCL, or Adaptive Monte Carlo Localization, is a probabilistic localization system that uses a particle filter to track the robot's pose against a known map. It scatters “particles” representing possible locations and, as the robot moves and observes the environment with its LiDAR, it discards unlikely particles until they cluster around the robot's true position.

4.4.4 Autonomous Can Approach

To approach the can and sit behind it, manual tuning of the Nav2 parameters was required. Specifically, the path alignment critic was tuned to ensure the robot stayed oriented on the path line, and the goal alignment critic was adjusted to ensure it reached the goal precisely without overshooting. If Go1 moved past the goal, there was a chance it would knock the can over or step on it. In addition, when the can was detected and its position was found, a 20 cm vector had to be subtracted from the goal coordinates so that Go1 would sit behind the can and not on top of it. See section 4.2 for vector subtraction.

The movement from the time of can detection to the final approach was solely completed by Nav2, and the can's position was only known from the original detection time as shown in the Goal Transformation section 4.2. This is also known as dead reckoning, where the robot relies on its internal estimation of movement (odometry) to reach a target without continuous visual feedback from the object. This highlights the importance of the EKF and AMCL accuracy, as any drift during this “blind” approach would result in a failed pick-up or collision.

4.4.5 Communication

To ensure reliable communication between the Raspberry Pi 5, Raspberry Pi 4-B, and the Unitree Go1, a continuous UDP stream of commands is utilized. The Raspberry Pi 5 acts as the central hub, issuing **HighCmd** structures to the Go1 to manage low-level movement (explained further in Control, Section 4.5), while inter-processor communication between the Pi 5 and Pi 4-B is handled via **Google Protocol Buffers (ProtoBufs)**.

ProtoBufs are a language-neutral, platform-neutral, and extensible mechanism for serializing structured data. Unlike text-based formats such as JSON or XML, ProtoBufs are compiled into a compact binary format, making them highly efficient for high-speed robotics applications where bandwidth and latency are critical. This efficiency is achieved by defining the data structure once in a schema file (.proto).

Once defined, this message can be seamlessly encoded and decoded by any supported language, such as C++ or Python, provided both systems possess the same protobuf schema. This ensures strict data integrity across the various hardware components of the system.

A message, similar to a class structure, can be defined using any combination of variables. An example of a message our team sends is a **GoalPosition**. Defined as follows:

Listing 1: GoalPosition Message

```
syntax = "proto3";

message GoalPosition {
    float x = 1;          // Relative X movement
    float y = 2;          // Relative Y movement
    float theta = 3;      // Relative rotation in degrees
}
```

4.5 Control

GO1 is controlled via HighCMDs, which consist of a UDP stream of high-level commands sent to the robot's onboard controller. These commands translate cmd-vel-nav geometry messages from ROS2 into omnidirectional movement instructions that the hardware can execute.

The specific HighCMD values transmitted include the robot's target orientation and position, along with direct velocity control parameters. Each command includes configuration flags to define the robot's state: the **gaitType** was consistently set to "trot" for movement, while the **operating mode** was dynamically toggled between "standing" and "sitting" depending on which phase of the pick up sequence Go1 was in.

CONFIGURATION FLAGS		
uint8 mode	<i>Operating Mode</i>	
uint8 gaitType	<i>Trot / Climb</i>	
uint8 speedLevel	<i>L1 / L2 / L3</i>	
BODY ADJUSTMENTS		
float footRaise		<i>m</i>
float bodyHeight		<i>m</i>
ORIENTATION & POSITION		
float euler[3]	<i>R, P, Y (rad)</i>	
float position[2]	<i>X, Y (m)</i>	
VELOCITY CONTROL		
float velocity[2]		<i>VX, VY (m/s)</i>
float yawSpeed		<i>rad/s</i>

Figure 13: HighCmd Structure sent over UDP to control Go1.

5 Verification, Validation, and Testing

To ensure that the autonomous litter-collecting robot performs as intended and meets the outlined design requirements, a structured test plan has been developed. The validation defines system-level tests to confirm reliable integration in real-world environments. Validation focuses on navigation, perception, and manipulation performance, while testing standards establish safety, repeatability, and compliance with relevant robotic guidelines. Together, these measures provide a rigorous framework for demonstrating the robot's functionality and readiness for deployment.

5.1 Test Plan

- **Navigation and Control:** The navigation system validation will proceed in a tiered approach, beginning with verifying communication through the Unitree SDK and confirming basic hardware control. Following this, the ROS Navigation Stack will be integrated and debugged by monitoring topic outputs with manually tuned inputs before any hardware deployment. The efficacy of the system will then be measured by sending goal positions to Nav2 and using physical tape markers at various coordinates and angles to benchmark accuracy. Finally, the process will conclude with comprehensive safety and stress evaluations, specifically testing the seamless switching between manual and autonomous modes, monitoring communication delays, and ensuring the reliability of emergency stop functionality.
- **Object Detection and Recognition:** Computer vision classifications on litter items of various sizes, shapes, and materials (e.g., plastic bottles, cans) in various orientations and scenes will be compared to ground truth in real-world tests and through model validation data sets. Physical properties observed will also be compared.
- **Manipulation:** In isolated experiments where only the manipulation subsystem is active, the gripper/end-effector will perform the Pick Up function to securely pick up and deposit detected litter items into the storage bin without dropping or damaging them.

5.2 Testing Standards

To ensure system reliability and safety, the testing framework employs a modular progression, beginning with software unit testing, integration testing, and ROS/Gazebo simulations prior to any physical hardware deployment. During physical trials, the system’s safety and kinematic performance are rigorously validated against ISO 13482:2014 and ISO 9283 standards, alongside hardware-level constraints such as capping the end-effector grasping force at a safe maximum of 160 N [10]. Finally, each isolated subsystem is evaluated against strict internal benchmarks for detection accuracy, navigation error, and collection efficiency before being cleared for the culminative field test.

5.3 Test Results

5.3.1 Computer Vision

The functional requirement **CV-01** for the computer vision subsystem aimed to maximize the model’s correct labeling of waste. Model performance was evaluated on the held-out test set of 429 images following training.

The model achieved a mask precision of 0.993, recall of 0.983, and mAP50 of 0.989, achieving a satisfactory threshold. Per-class performance was similarly strong, with 288 correct can detections and 242 correct bottle detections in the test set, and false negative rates of only 0.01 and 0.02 for can and bottle, respectively. The confusion matrices in Figure 14 provide per-class breakdowns. The raw counts confirm that false positives on background regions were low in absolute terms, where 10 background regions were misclassified as can and 8 as bottle. The normalized matrix exaggerates this due to the small number of true background samples in the test set, and should be interpreted alongside the raw counts. The training and validation loss curves in Figure 15 confirm smooth convergence with no signs of overfitting.

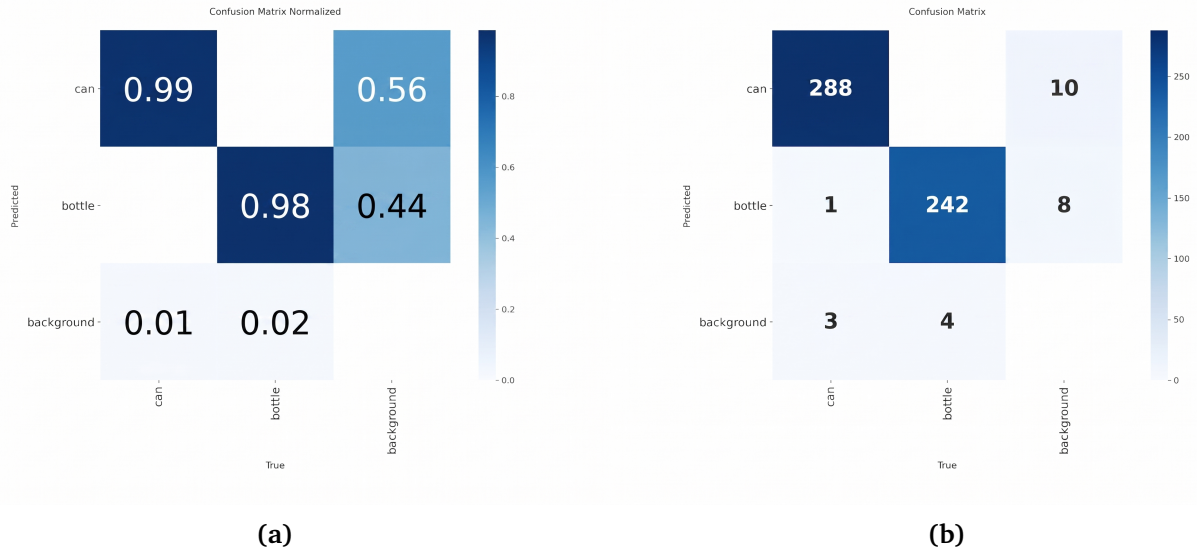


Figure 14: (a) Normalized confusion matrix evaluated on the held-out test set. Rows represent predicted classes, columns represent true classes. (b) Raw confusion matrix evaluated on the held-out test set.

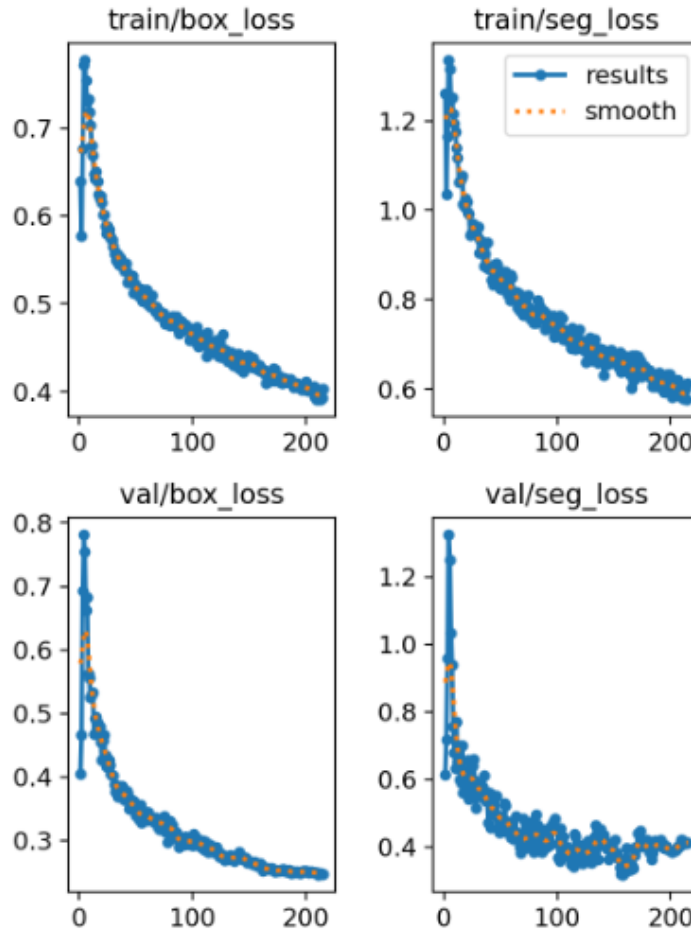


Figure 15: Training and validation box and segmentation loss curves. Smooth convergence with losses tracking closely indicates no overfitting.

Requirement **CV-02**, of minimizing physical property estimation was validated against a standard 355 ml aluminum can, which has a known height of 12.3 cm and body diameter of 6.6 cm. The inference pipeline reported dimensions of 14.0 cm $\times$ 7.2 cm, corresponding to errors of 14.1% and 9.1% for the major and minor axes, respectively, and both were within the specified threshold. The consistent overestimation is attributable to ellipse fitting tracing the outer boundary of the segmentation mask rather than the object’s true geometry. An example inference output including the reported dimensions is shown in Figure 16 below.



Figure 16: Example inference output showing instance segmentation mask, estimated dimensions, distance, and orientation angle for a 355 ml can.

The measured angle of the can had an approximate relative error ± 20 degrees with respect to ground truth in all trials, but was highly dependent on the perspective of the camera. Future iterations, should collect multiple view-points to consider the can angle θ_C .

Qualitatively, indoor performance was reliable under controlled lighting conditions. Occasional false positives were observed on seemingly random objects with no obvious visual resemblance to cans or bottles, suggesting the model occasionally activates on texture or context cues rather than true object geometry. The model also intermittently lost track of cans at certain viewing angles, where the aspect ratio or occlusion of the object caused the segmentation mask to drop below the confidence threshold.

Outdoor performance was more sensitive to lighting conditions. Under direct sunlight, detection quality degraded significantly, and reliable results could only be obtained by artificially shading the area within the camera’s field of view. This highlights a key limitation of training exclusively on internet-sourced datasets, which do not adequately represent high-contrast outdoor lighting conditions. Future work should incorporate domain-specific training data collected in the intended deployment environment, or employ additional image processing to compensate for varying lighting conditions.

Inference speed was measured on the Raspberry Pi to validate the hardware deployment constraint. The nano variant achieved 6–7 fps on the Pi, while the small variant achieved 1–2 fps, which confirmed that direct on-device inference at a practical frame rate was not feasible with the small model, and motivating the offload of the vision pipeline to an external computer.

Overall, the computer vision subsystem met its functional requirement **CV-01**, maximizing mAP during several tests. The identified failure modes are consistent with known limitations of training on internet-sourced datasets and are expected to improve with domain-specific data collection in outdoor litter environments.

5.3.2 Manipulation

To test the success rate of manipulator-isolated collections, a can was randomly placed within the robot’s reachable workspace (including position and orientation), and the autonomous collection process was deployed 15 times. Out of the 15 trials, 13 trials were successful, resulting in a 86.67% mission success rate, given an ideal approach towards the can. It was seen that the robot would often fail on cases where the can was slightly out of reach or when the environmental conditions would not allow the computer vision model to detect the can in certain positions. Ultimately, the manipulator met the team’s determined requirements (**ME-01-03**) as it reliably succeeded in its individual test of autonomous detection and collection.

5.3.3 Navigation and Control

Hardware Communication: Connection via the Unitree SDK was successfully verified by publishing twist commands to the HighCMD topic using a teleoperation node. Go1 demonstrated responsive omnidirectional movement consistent with native remote control performance.

Controller Tuning: Initial autonomous trials showed a tendency for the robot to overshoot goals and oscillate off-path. This was resolved by iteratively tuning the Nav2 Controller Server parameters: the GoalAlignmentCritic was increased to dampen approach speed and prevent overshoot, while the PathAlignmentCritic was weighted more heavily to maintain strict heading alignment with the global path. This allowed Go1 to approach the can from behind.

Coordinate Frame Correction: A systematic 20 cm steady-state error was identified during goal

approach. Investigation revealed a transform error: the navigation goal was being referenced to the base-link (center of body) rather than the end-effector/sensor link at the front of Go1. By correctly transforming the “can” vector to the front-mounted manipulator and camera frame, the offset was eliminated, resulting in precise alignment and satisfying **NA-01**.

Safety and Latency: E-stop functionality was verified to halt all motor power as part of requirement **CO-04**. When the controller was actuated, it would overwrite the Nav2 velocity commands.

5.3.4 Field Test

A preliminary field test of the system was conducted on April 6th, 2026, near the Summerhill building on Queen’s University campus (See Figure 1). The site featured non-planar grass terrain and was selected as there was a large volume of litter present including aluminum cans and plastic bottles. Testing occurred under direct sunlight, providing a rigorous challenge for the perception subsystem and field validation of **CV-01**.

The field test partially satisfied requirement **OP-04** with three modifications to the collection strategy described in Section 3.2. First, the timing of the field trial and initial robot position were carefully selected to ensure safe operating conditions and improve the likelihood of a successful initial search. Second, outdoor sun light affected object detection performance inducing sensor saturation and high-contrast shadows on reflective waste surfaces. To combat this, artificial shadowing was applied to target objects to prevent bounding box instability. Third, locomotion was performed manually, replacing the Velocity Commander shown in Figure 7. After the goal transform was completed, the operator teleoperated the robot to approximately the goal position and manually sent the Posture stabilized flag. All other operations remained unmodified.



Figure 17: Cans and bottles collected during the field trial at Summerhill, April 2026.

Under these conditions, the robot achieved a 37.5% autonomous retrieval success rate during eight trials, successfully collecting two cans and one bottle. The remaining five trials revealed a consistent failure mode characterized by pose-dependent detection loss. While the CV system successfully initialized tracking from a distance, transitions during the Waste Scanning and Pick and Place state poses often resulted in false negative detections. During the field trial, the team ensured requirement **CO-03** was satisfied throughout as safety was of critical importance in the field. Requirement **OP-03** was also successfully met during the test as the software and electrical subsystems performed flawlessly for approximately 40 minutes.

6 Compliance with Specifications

Table 1 summarises the compliance status of each requirement defined in Appendix A against the results presented in Section 5.3.

Table 1: Compliance summary mapped to requirements defined in Appendix A.

ID	Subsystem	Status	Evidence
CV-01	Vision	Met	Model achieved mAP50 of 0.989, mask precision of 0.993, and recall of 0.983 on the held-out test set. Performance degraded under direct sunlight outdoors, requiring artificial shading during field testing.
CV-02	Vision	Met	Dimension estimation errors of 14.1% (major axis) and 9.1% (minor axis) were measured against a standard 355 ml can. Orientation estimation carried an approximate relative error of $\pm 20^\circ$ depending on camera perspective.
ME-01	Manipulator	Met	13 of 15 isolated manipulation trials resulted in successful grasps, yielding an 86.7% success rate across varied object positions and orientations within the reachable workspace.
ME-02	Manipulator	Met	No collisions with the ground or chassis were observed during manipulation trials. The wrist tilt extension feature ensured clearance was maintained even at full arm extension.
ME-03	Manipulator	Met	Trajectories were observed to be smooth and stable during isolated testing. The local search recovery algorithm successfully resumed closed-loop control after object loss without erratic motion, although it could take a great deal of time to finalize the approach occasionally.
OP-01	Operations	Met	All subsystem and integration testing was conducted indoors on hard flat surfaces under controlled lighting, with no environmental failures recorded.
OP-02	Operations	Met	SLAM successfully generated an occupancy map of the showcase environment (Mitchell Hall Room 235, see Figure 12). AMCL localization within the map was stable throughout navigation trials.
OP-03	Operations	Met	Electrical and software systems operated continuously for approximately 40 minutes during field testing without failure.

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Table 1 continued from previous page

ID	Subsystem	Status	Evidence
OP-04	Operations	Partially Met	Full autonomous collection was demonstrated indoors. During the outdoor field test, manual teleoperation replaced the autonomous locomotion stage due to a lack of vertical features for SLAM navigation.
CO-01	Admin	Met	Total expensed budget was \$425 CAD, within the \$500 constraint (see Appendix B).
CO-02	Admin	Met	All project deliverables were submitted by the required deadlines.
CO-03	Admin & Safety	Met	A team member supervised all hardware testing and field trials, with manual intervention available at all times.
CO-04	Safety	Partially Met	Software termination functions were implemented however due to the distributed system architecture, a physical E-stop was not implemented in the prototype.
NA-01	Navigation	Met	After tuning Nav2 controller parameters and correcting the goal frame transform, Go1 reliably converged to target positions during indoor navigation trials with no steady-state offset.

7 Professional, Ethical, and Societal Considerations

The deployment of an *Autonomous Small Waste Collection Robot* in public spaces demands rigorous attention to professional responsibility, ethical integrity, and societal impact. Professionally, the system must meet high standards of safety, reliability, and robustness under diverse and unpredictable real-world conditions. This includes systematic validation across edge cases involving human interaction, dynamic obstacles, and environmental variability. Transparency regarding system capabilities and limitations is essential to prevent over reliance or misuse. Equally important is life-cycle accountability with clear assignment of responsibility for developers, operators, and municipalities must be established in the event of malfunction, harm, or service failure. Economic viability should be evaluated not only in terms of upfront cost, but also considering a full life-cycle encompassing maintenance, integration with existing public infrastructure, and disposal at end-of-life.

From a societal perspective, such this project can improve municipal efficiency and reduce human exposure to hazardous waste, but they may also contribute to job displacement, particularly for low-wage sanitation workers. Correspondence with the City of Kingston suggests direct displacement is limited as most work in the scope of this project are tackled by seasonal or volunteer-based litter collection [1]. Nevertheless, responsible deployment should consider workforce transition strategies, such as retraining or augmenting human roles rather than fully replacing them.

Additionally, public acceptance and trust are essential. Robots operating in shared spaces must

behave predictably, respect social norms, and integrate smoothly into communities. Privacy concerns of the robot using sensor data to collect sensitive information about individuals or private property must be addressed it should be clearly communicated what data is being captured and how it is used. There is also potential for concerns around bias and fairness particularly if the robot’s deployment could reinforce existing inequalities in municipal services (e.g., prioritizing work in certain neighborhoods over others).

We must also consider the safety of the autonomous system itself. There are inherent risks to operating in unknown outdoor environments, where the robot could get stuck or damaged. Furthermore, “robot bullying” pertaining to acts of vandalism and damage to public service robots by humans with malicious intent is a significant risk during deployment [11]. For early adoption of the *Autonomous Small Waste Collection Robot*, we propose a human supervisor works alongside the robot as an extension of **CO-03** to mitigate these risks and assist with tasks outside the robot’s operational bounds.

Ultimately, the responsible integration of such systems requires continuous engagement with stakeholders, including municipalities, workers, and the public. Notably, the *Autonomous Small Waste Collection Robot* has already demonstrated a positive local impact through the removal of eight pollutants from Summerhill during field trials. Building on this early outcome, future deployment should aim to extend these benefits in a measured and responsible manner.

8 Conclusions and Reflections

The *Autonomous Small Waste Collection Robot* implements a modular robotic system capable of autonomously detecting, approaching, and collecting cans and bottles. The solution successfully defines a complete hardware and software architecture accustomed for indoor use.

To validate system performance, a structured test plan was developed and validated, emphasizing navigation accuracy, perception reliability, and manipulation effectiveness. Testing protocols adhere to recognized robotics standards, ensuring safety, repeatability, and compliance with best practices. This project aims to demonstrate the feasibility of autonomous litter collection as a practical, deployable solution to the prevalent litter problem. Through rigorous testing, it was determined that each subsystem successfully met the predefined goals, indicating overall reliability of each independent component.

Initial deployment of the robot prototype shows promising results through successful mission completion in a majority of test trials, though due to limited full-system tests, these results can not be labeled as statistically significant. It is also to be acknowledged that most tests were completed in an indoor controlled environment, with idealized conditions and consistent test setups pertaining to physical environment, lighting conditions, and lack of disturbances.

Through this initial development on the *Autonomous Small Waste Collection Robot*, it was learned that integration of the complete system should be prioritized over individual success rates, as many unforeseen obstacles are revealed through full-system testing. For future developments of the project, the transition to outdoor use with GPS-enabled localization can be explored, allowing the system to be deployed in urban environments where litter is more prevalent.

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A System Requirements

The detailed design parameters, performance objectives, and constraints that governed the development of the *Autonomous Small Waste Collection Robot* are detailed in Table 2. Where empirical baselines were unavailable, performance metrics were defined as optimization objectives to validate successful behavior.

Table 2: Project requirements justified and classified by subsystem.

ID	Subsystem	Requirement	Objective / Metric	Justification
CV-01	Vision	Identify and locate target waste within the camera frame at various depths.	Maximize mean average precision across diverse lighting and scenes with direct line of sight to the target.	Ensures the robot can reliably detect the target during search and pick up sequences.
CV-02	Vision	Estimate physical properties (dimensions, orientation).	Minimize estimation error relative to ground truth.	Necessary to calculate safe approach vectors and ensure the gripper width is sufficient prior to contact.
ME-01	Manipulator	Securely grasp target waste (bottles and cans < 300 g, < 1 L).	Maximize successful grasp rate across diverse object orientations and geometries.	Payload limits are dictated by the manipulator’s motor torque and end-effector specifications.
ME-02	Manipulator	Execute collision-free trajectory paths relative to known obstacles.	Zero collisions with the ground or mobile platform chassis.	Protects both the manipulator hardware and the expensive quadruped base from self-inflicted damage during operation.
ME-03	Manipulator	Execute smooth and stable waypoint transitions.	Minimize absolute kinematic tracking error and overshoot.	Prevents dropping collected waste during transport and reduces instability on the quadruped base.
OP-01	Operations	Operate in controlled environments (dry, 5–30°C, hard surfaces).	Zero environmental points of failure during testing.	Protects components from environmental damage and isolates variables for testing during initial proof-of-concept development.

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Table 2 continued from previous page

ID	Subsystem	Requirement	Objective	Justification
OP-02	Operations	Operation on planar surface with distinct vertical features.	Out-of-box SLAM algorithms can be deployed on the system and achieve expected results.	2D LiDAR based SLAM relies on the sensor staying in one plane and sufficient reliable observations.
OP-03	Operations	Electrical and software systems must be robust and reliable.	Power system will provide necessary power to all components for proper operation and maximize battery life.	Operation is limited by available energy and dependent on proper software implementation.
OP-04	Operations	Under OP-02 and OP-03 the robot can perform the collection process described in Section 3.2 autonomously.	The system is capable of collecting waste without human intervention.	A core requirement of this project, this demonstrates autonomy of the project.
CO-01	Admin	Adhere to an allocated project budget.	Total bill of materials (expensed) $\leq$ \$500.	Strict constraint dictated by course instructions and available funding.
CO-02	Admin	Meet the final academic delivery deadline.	Adhere to all delivery deadlines provided by course instructors.	Required constraint for project evaluation and grading.
CO-03	Admin & Safety	The physical system must be supervised by a team member.	An operator will be prepared for manual intervention at all times during active development and testing.	A human supervisor is a risk mitigation strategy to avoid external or self-inflicted damage to the robot.
CO-04	Safety	Implement hardware and software emergency stops.	System shuts down within < 500 ms of activation.	Ensures compliance with standard service robotics safety guidelines and protects bystanders and the robot itself.
NA-01	Navigation	Converge to a desired state given a navigational goal.	Minimize error of final state to a reference state from an initial state and minimize time of arrival.	Necessary for the robot to traverse its environments and approach targets.

B Bill of Materials

Table 3 lists the main components purchased and used in the *Autonomous Small Waste Collection Robot* project. Prices are rounded to the nearest five dollars (CAD) and include shipping if not stated otherwise.

Table 3: Bill of Materials

Item Category	Description	Vendor/Source	Qty	Cost (CAD)	
Mobile Platform	Unitree Go1 Edu	Ingenuity Labs	1	\$8,500	[12] [*]
Manipulator	OpenMANIPULATOR-X	Ingenuity Labs	1	\$550	[8] [*]
Hardware	Waste Bin	Amazon	1	\$5	[13]
	Mounting components (nuts, bolts, adhesive)	Amazon	20	\$20	[14]
	Raspberry Pi 4B	Personal	1	\$75	[15] [*]
	Raspberry Pi 5	ESP Lab	1	\$145	[16] [*]
	Intel RealSense RGB-D Camera	Offroad Robotics	1	\$300	[9] [*]
	RPLiDAR C1	Slamtec	1	\$120	[7]
	MPU6050 IMU	Amazon	1	\$10	[17]
	Electrical Connectors	DigiKey	1	\$30	[18]
	U2D2 Power Hub Board	eBay	1	\$60	[19]
	SMPS 12V 5A Battery	RobotShop	1	\$45	[20]
	U2D2 Board	RobotShop	1	\$55	[21]
	Filament	Personal	2	\$40	[22] [*]
	2mm Rods and Bearings	McMaster-Carr	1 set	\$35	[23]
Subtotal				\$10,035	
TOTAL BUDGET				\$10,035	

^{*} Items from “Ingenuity Labs”, “ESP Lab”, “Offroad Robotics”, and “Personal” were provided to the project free of charge. The total budget excluding these items was \$425, compliant with requirement **CO-01**.

Author Contributions

Nathan Duncan: Methodology, Software, Visualization, Validation, Writing – Original Draft.

Daniel Poon: System Architecture, Software, Validation, Writing – Original Draft, Conclusion

Daniel Dubinko: Methodology, Software, Sensors and Components, Validation, Writing – Original Draft

Connall Milberry: Software, Validation, Writing – Original Draft, Compliance with Specifications